

Identifying Conics from Their Equations

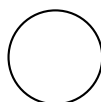
Naming a conic from its equation, then reading off a circle's center and radius, a parabola's vertex and axis, and an ellipse's vertices

How to use this sheet.

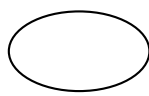
- Every conic can be written in the *general form* $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$. On this sheet $B = 0$ every time, so the curve's axes line up with the grid.
- The *indicator* $B^2 - 4AC$ decides the family before you do any other work. Read the model below, then use it on every naming problem.
- Once you know the family, the standard form gives you the key features. Show the completing-the-square work — the answer blank is for the finished feature.

Concept model: the indicator sorts the four families

indicator $B^2 - 4AC$	extra condition	family
negative	$A = C$	circle
negative	$A \neq C$	ellipse
zero	only one square term	parabola
positive	squares of opposite sign	hyperbola



circle



ellipse



parabola



hyperbola

How every family looks when it is centred and lined up with the axes. No values shown — use the numbers given in each problem.

1. Read the coefficients, then name it. $x^2 + y^2 - 6x + 4y - 12 = 0$

Match the equation against $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$ and fill the table.

A	B	C	$B^2 - 4AC$

Indicator = _____ Is $A = C$? _____ Family: _____

2. Same model, different signs. $9x^2 - 4y^2 = 36$

Move everything to one side first, then read A , B and C .

A	B	C	$B^2 - 4AC$

Indicator = _____ Family: _____

3. A missing square. $y = x^2 - 6x + 5$

Rewrite it as $x^2 - 6x - y + 5 = 0$ so that the general form is visible. What are A , B and C now? Notice what happens to the indicator when one of the squares is missing.

$A =$ _____ $B =$ _____ $C =$ _____ Indicator = _____

Family: _____

4. Circle: find the center and the radius. $x^2 + y^2 - 6x + 4y - 12 = 0$

This is problem 1's equation again, now that you know it is a circle. Group the x terms and the y terms, complete both squares, and move the constant across:

$$(x - h)^2 + (y - k)^2 = r^2 \quad \text{with} \quad h = -\frac{D}{2}, \quad k = -\frac{E}{2}, \quad r = \sqrt{h^2 + k^2 + 12}$$

$h =$ _____ $k =$ _____ $r =$ _____

Standard form: _____

5. Parabola: vertex, axis, and where it crosses. $y = x^2 - 6x + 5$

Three key features, in the order they come out of the equation.

axis of symmetry $x = -\frac{b}{2a}$	$x =$ _____
vertex y -value: put that x back in	$y =$ _____
x -intercepts: solve $x^2 - 6x + 5 = 0$	$x =$ _____

Vertex = (_____, _____) Does it open up or down? _____

6. Ellipse: how far out in each direction? $\frac{x^2}{25} + \frac{y^2}{9} = 1$

In this standard form the two denominators are a^2 and b^2 , and the focal distance satisfies $c^2 = a^2 - b^2$.

$a =$ _____ $b =$ _____ $c =$ _____

Vertices: $(\pm$ _____, $0)$ Co-vertices: $(0, \pm$ _____) Foci: $(\pm$ _____, $0)$

7. Two squares, same sign, different sizes. $x^2 + 4y^2 + 2x - 24y + 33 = 0$

Compute the indicator first, then use the extra condition in the concept model to decide between the two families a negative indicator allows.

Indicator = _____ $A =$ _____ $C =$ _____

Family: _____ Why not the other one? _____

8. Careful with this one. $x^2 + y^2 + 10x - 4y + 13 = 0$

Complete both squares. Then read the standard form carefully: the right-hand side is r^2 , not r .

Center = (_____, _____) $r^2 =$ _____ $r =$ _____

9. A nonlinear model. A ball is thrown from the ground and its path over level ground follows $y = -x^2 + 6x$, where x is the horizontal distance in metres and y is the height in metres.

The highest point of any parabola sits on its axis of symmetry, and here that axis is the line $x = 3$. So the highest point is the single solution of the system

$$y = -x^2 + 6x \quad \text{and} \quad x = 3.$$

Substitute and solve. Then say in one sentence what each coordinate of the answer means for the ball.

Highest point = (_____, _____)

10. Explain the difference. Mia looks at $4x^2 + 9y^2 = 36$ and $4x^2 - 9y^2 = 36$ and says: “Both have an x square and a y square, so both are ellipses.”

She is right about one of them. Compute the indicator for each equation, then write two or three sentences explaining what in the equations makes them different curves — and what Mia was looking at instead.
